

ADVANCED COMPUTER NETWORKS
(MTITE101)

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Practical 2:

1. Aim: To capture and analyze IPv4 and IPv6 packets using Wireshark and study their respective protocol header fields.

2. Commands Run

IPv4:

ping -c 4 8.8.8.8

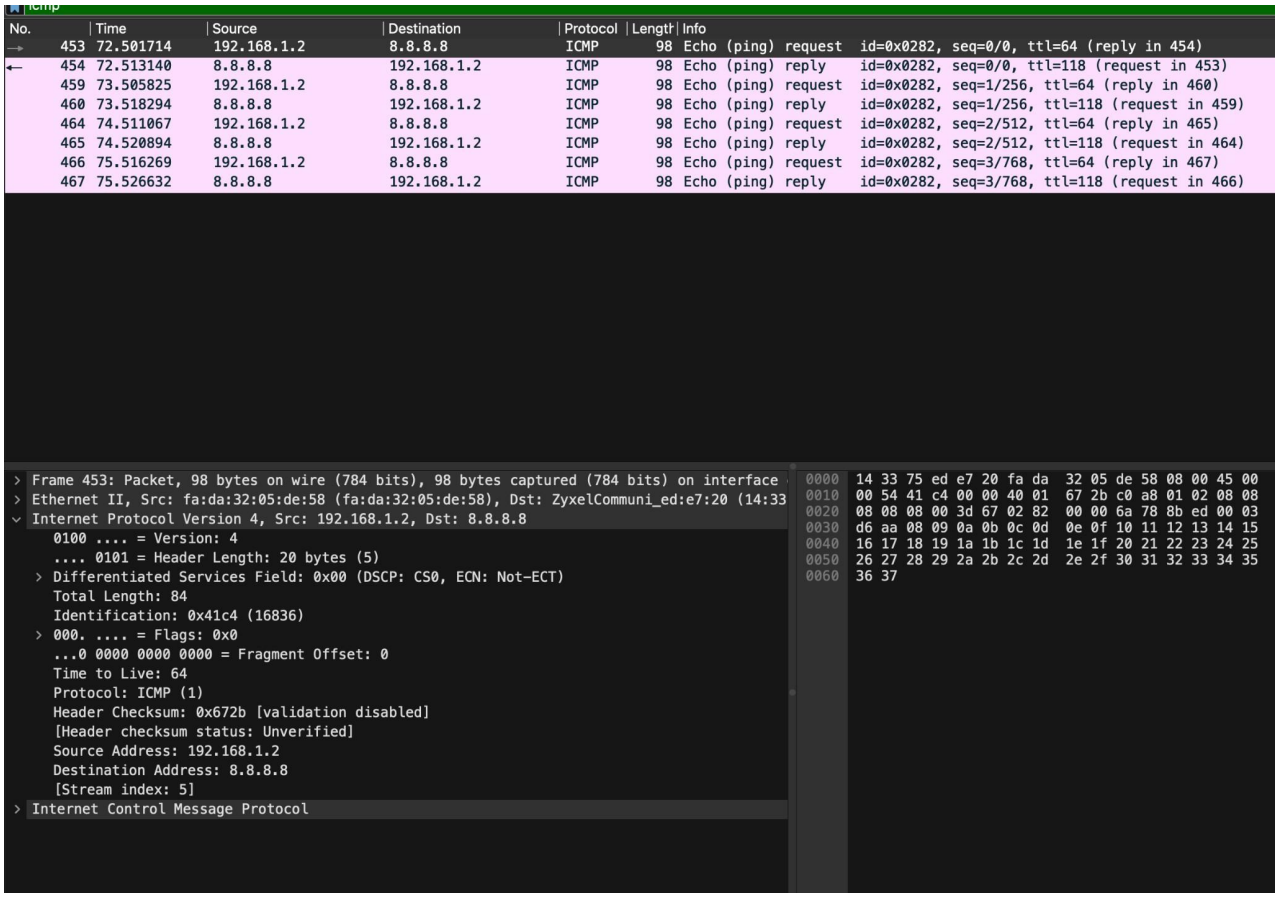
Function: Generates ICMP-based IPv4 traffic for packet capture and analysis in Wireshark.

When you run ping -c 4 8.8.8.8

Echo Request generated

- Your OS builds an ICMP Echo Request message:
  - Type = 8 (Echo Request)
  - Code = 0
  - Identifier + Sequence number (to match replies)
  - Payload

Output:


icmp
No. Time Source Destination Protocol Length Info
453 72.501714 192.168.1.2 8.8.8.8 ICMP 98 Echo (ping) request id=0x0282, seq=0/0, ttl=64 (reply in 454)
454 72.513140 8.8.8.8 192.168.1.2 ICMP 98 Echo (ping) reply id=0x0282, seq=0/0, ttl=118 (request in 453)
459 73.505825 192.168.1.2 8.8.8.8 ICMP 98 Echo (ping) request id=0x0282, seq=1/256, ttl=64 (reply in 460)
460 73.518294 8.8.8.8 192.168.1.2 ICMP 98 Echo (ping) reply id=0x0282, seq=1/256, ttl=118 (request in 459)
464 74.511067 192.168.1.2 8.8.8.8 ICMP 98 Echo (ping) request id=0x0282, seq=2/512, ttl=64 (reply in 465)
465 74.520894 8.8.8.8 192.168.1.2 ICMP 98 Echo (ping) reply id=0x0282, seq=2/512, ttl=118 (request in 464)
466 75.516269 192.168.1.2 8.8.8.8 ICMP 98 Echo (ping) request id=0x0282, seq=3/768, ttl=64 (reply in 467)
467 75.526632 8.8.8.8 192.168.1.2 ICMP 98 Echo (ping) reply id=0x0282, seq=3/768, ttl=118 (request in 466)
Frame 453: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface
Ethernet II, Src: fa:da:32:05:de:58 (fa:da:32:05:de:58), Dst: ZyxelCommuni_ed:e7:20 (14:33
Internet Protocol Version 4, Src: 192.168.1.2, Dst: 8.8.8.8
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0x41c4 (16836)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0x672b [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.2
Destination Address: 8.8.8.8
[Stream index: 5]
Internet Control Message Protocol
0000 14 33 75 ed e7 20 fa da 32 05 de 58 08 00 45 00
0010 00 54 41 c4 00 00 40 01 67 2b c0 a8 01 02 08 08
0020 08 08 08 00 3d 67 02 82 00 00 6a 78 8b ed 00 03
0030 d6 aa 08 09 0a 0b 0c 0d 0e 0f 10 11 12 13 14 15
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
0060 36 37

What Wireshark shows with icmp filter

Ethernet II header → Source MAC (your NIC) → Destination MAC (router/next hop).

IPv4 header → Source IP (10.100.167.35) → Destination IP (8.8.8.8).

ICMP header → Echo Request or Echo Reply, with type/code, checksum, identifiers.

Payload → The "data" section (often 32 bytes by default). In your capture, you saw ASCII like abcdef ghijklmn... — that's the ping payload.

IPv6:

ping6 -c 4 ::1

Function: Generates ICMPv6 traffic using the IPv6 loopback address for packet capture and analysis in Wireshark.

Your host generates **ICMPv6 Echo Requests** (Type = 128).

- Destination = ::1 (your own loopback).
- Source = ::1 as well.
- Payload = filler data (like abcdef...).

The packets never leave your NIC — they're handled internally by the OS.

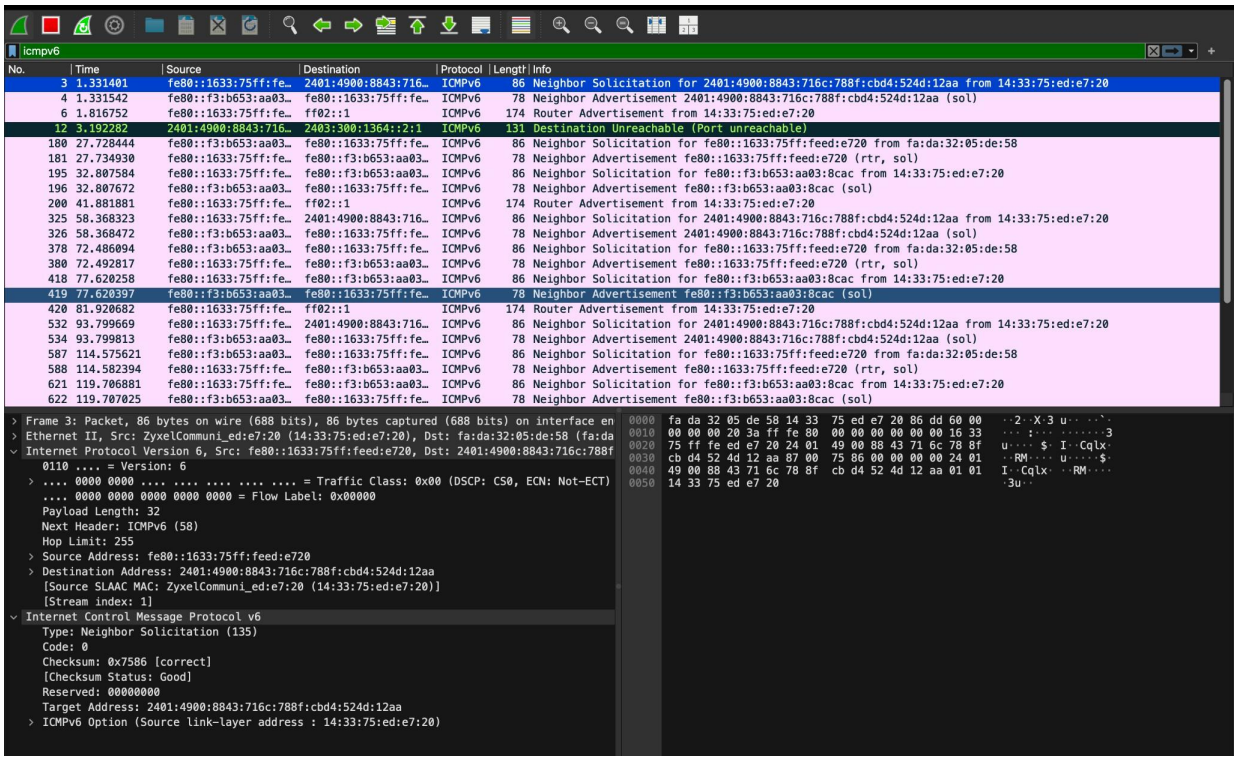
- No Ethernet frame is sent out, because loopback is a virtual interface.

The OS immediately replies with **ICMPv6 Echo Replies** (Type = 129).

- Same identifier and sequence number.
- Same payload.

ping6 prints the results: round trip times are extremely low (usually <1 ms), because the packets never leave your machine.

Output:



3. Result: IPv4 and IPv6 packets were successfully captured and analyzed using Wireshark. The corresponding IPv4 and IPv6 header fields were observed and studied.